Unresolved companion detection and spectral extraction

Hyperspectral speckle imaging uses multi-frame myopic deconvolution to recover wavelength-dependent, nearly diffraction-limited imagery of resident space objects from broadband short-exposure data and wavefront-sensor measurements (Hall & Jefferies 2022; Johns et al. 2024). Each data frame integrates light over the visible/NIR bandpass, from approximately 400 to 1,000 nm. Spectral information is encoded through the wavelength dependence of the point-spread function (PSF) due to atmospheric turbulence and instrumental diffraction. Recovering this information therefore requires accurate PSF estimates across the bandpass.

Here, we report current progress toward detecting and characterizing faint unresolved companions near brighter resolved satellites. Using simulated hyperspectral speckle data, we developed and tested methods for companion detection and spectral estimation. Such objects could represent debris or other uncooperative objects.

The results establish feasibility under the modeled observing conditions, but they do not yet establish performance on on-sky data or across the full range of PSF uncertainty, primary morphology, and primary–companion separation. The next phase will test these methods on suitable observations and determine whether the simulated gains persist under operational conditions.

Figure 1 shows reconstructed images of a simulated $m_v = +6$ satellite with an unresolved $m_v = +13.1$ companion[1]. The wavelength-averaged reconstruction shows a possible feature at the 3 o'clock position, distinct from the restoration noise elsewhere. However, the feature appears in only some recovered wavelength slices, so the reconstruction alone cannot establish a detection. We therefore develop a quantitative detection statistic and assess its detection probability.

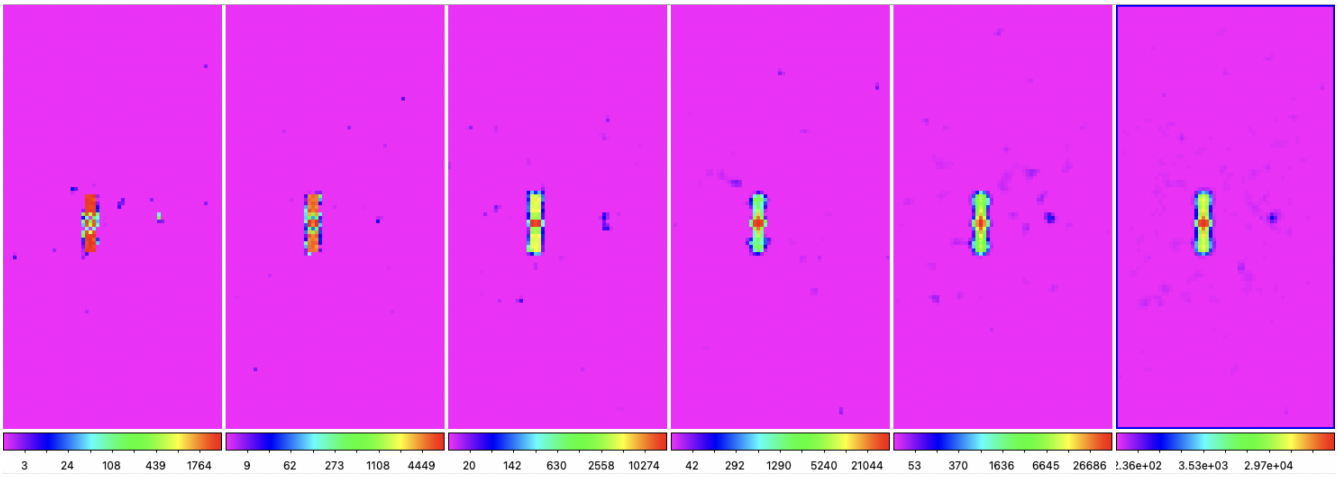


Figure 1: Reconstructed images for an $m_v = +6$ satellite with an unresolved $m_v = +13.1$ companion, using 100 frames of broadband speckle data from 400 to 915 nm with 5-ms exposures. Data were acquired with a 3.65-m telescope through atmospheric turbulence with $r_0 = 17$ cm at 500 nm. Left to right: recovered object slices at 400, 457, 533, 640, and 800 nm; far right: wavelength-averaged reconstruction over all 10 recovered slices. Images use logarithmic scaling and a rainbow color table to display their large dynamic range.

Matched filter as a detector

The restoration algorithm uses weighted least squares and the quasi-Newton optimizer L-BFGS-B (Appendix A.1). At the end of the optimization, the weighted residuals are

$$r_k(\vec{x}) = \frac{d_k(\vec{x}) - m_k(\vec{x})}{\sqrt{m_k(\vec{x}) + n^2(\vec{x})}}, \tag{1}$$

where $d_k(\vec{x})$ is the kth data frame and

$$m_k(\vec{x}) = \sum_{\lambda} o_\lambda(\vec{x}) \odot h_{\lambda,k}(\vec{x}), \tag{2}$$

is the corresponding model. Here, o_λ is the object at wavelength λ, $h_{\lambda,k}$ is the PSF for frame k, and $\odot$ denotes convolution. Under the model, the residuals are white with zero mean and unit variance.

We model the object as the sum of a resolved primary satellite and an unresolved companion,

$$o_\lambda(\vec{x}) = s_\lambda(\vec{x}) + \alpha(\lambda)\delta(\vec{x} - \vec{x}_0), \tag{3}$$

[1]An example of an $m_v \simeq +6$ satellite is Iridium NEXT at an orbital altitude of approximately 780 km. It has an area of approximately 12 m^2. An $m_v = +13.1$ object with the same reflectivity profile as the primary would have dimensions of roughly 13×13 cm. We have not modeled the spatial extent of either object, only their brightness difference.

where $s_\lambda(\vec{x})$ is the primary signal and $\delta(\vec{x}-\vec{x}_0)$ is a delta function at the companion location. Because the companion is faint relative to the primary, we subtract the primary-only model to form

$$r'_k(\vec{x}) = d_k(\vec{x}) - m'_k(\vec{x}), \tag{4}$$

with

$$m'_k(\vec{x}) = \sum_\lambda s_\lambda(\vec{x}) \odot h_{\lambda,k}(\vec{x}). \tag{5}$$

These residuals are also modeled as Gaussian with zero mean and unit variance.

We correlate the primary-subtracted residual in each frame with that frame's wavelength-averaged PSF and co-add the correlations:

$$MF(\vec{x}) = \sum_k r'_k(\vec{x}) \star \bar{h}_k(\vec{x}), \tag{6}$$

where

$$\bar{h}_k(\vec{x}) = \frac{1}{n_\lambda} \sum_\lambda h_{\lambda,k}(\vec{x}), \tag{7}$$

n_λ is the number of recovered spectral slices, the PSFs have unit volume, and $\star$ denotes correlation. We use $n_\lambda = 10^2$. Equation 7 provides the practical detector template.

The resulting matched-filter map contains compact companion signal superposed on a broad, smooth pedestal. This pedestal is a seeing halo caused by imperfect primary subtraction; it is strongest near the primary and declines slowly with radius. We remove it with the high-pass filter

$$MF_0(\vec{x}) = MF(\vec{x}) - (MF \odot G)(\vec{x}), \tag{8}$$

where $G(\vec{x})$ is a Gaussian kernel with $\sigma = 3$ pixels. Primary subtraction removes the primary signal but not its shot noise, leaving an irreducible floor

$$\sigma_k^2(\vec{x}) \approx m_k(\vec{x}) + n^2(\vec{x}). \tag{9}$$

We estimate the radial noise profile about the primary center and use its reciprocal to weight $MF_0(\vec{x})$. The resulting SNR map has zero mean and unit variance. Peaks above approximately 5σ are treated as candidate detections.

Figure 2 shows the matched-filter map for the data in Figure 1. The companion produces a 4.95σ peak, effectively at the nominal 5σ candidate-detection threshold.

Figure 3 gives peak detection SNR as a function of companion magnitude for this observing scenario. With 100 frames at 5-ms sampling, the approximate detection limit is $m_v \leq +13.3$. Additional frames improve the SNR; for independent noise, the expected scaling is approximately $\sqrt{N_{\text{frames}}}$.

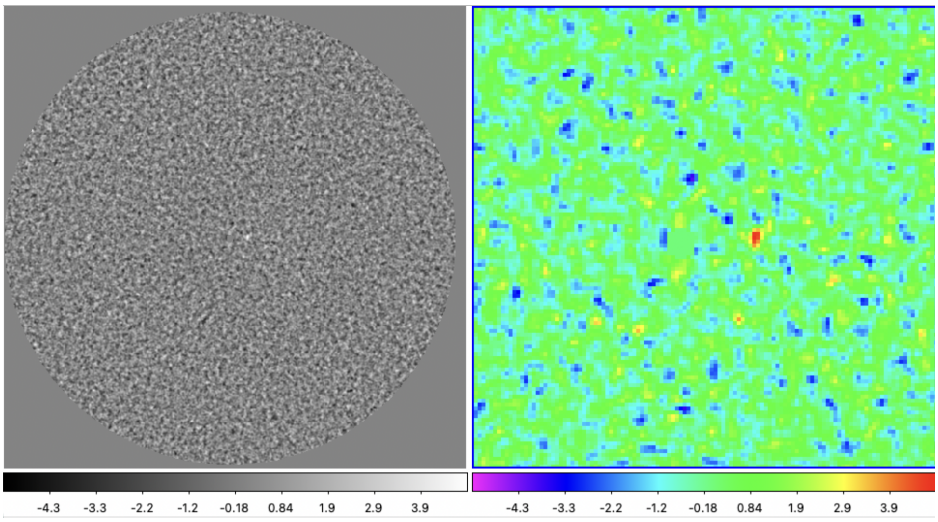


Figure 2: Weighted matched-filter SNR map for the data in Figure 1. Left: full map with linear scaling. Right: 4× enlargement of the central region. The color palette highlights the companion peak (red), which reaches 4.95σ.

[2]An SVD analysis of reflectance spectra for 47 materials commonly used in satellites, provided by E. Plis, found that 10 modes represent the spectra with spectral angle below 10°. We therefore use 10 recovered spectral slices.

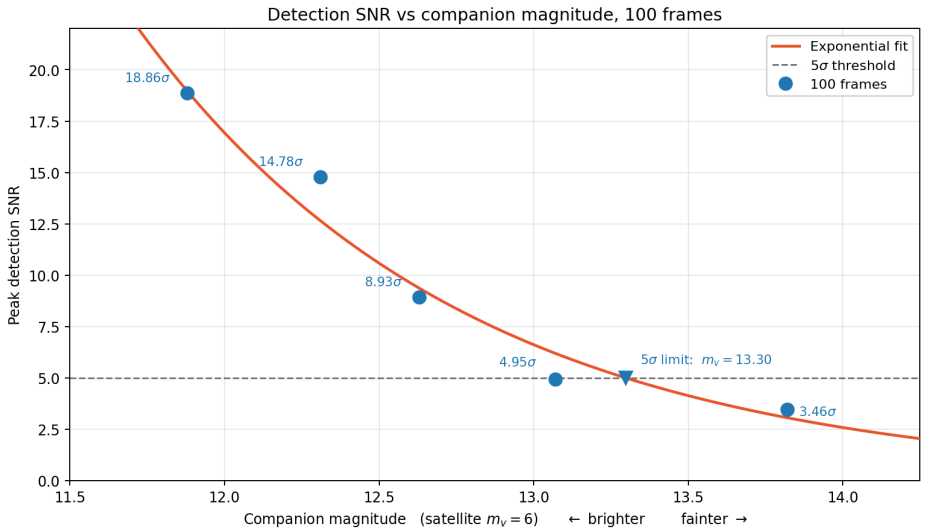


Figure 3: Peak detection SNR versus companion magnitude for an $m_v = +6$ primary using 100 frames. Blue circles show measured peak SNR from the radially whitened matched-filter co-add at a separation of 19.06 pixels. The orange curve is the least-squares fit SNR $= \exp(a + b\,m)$, with $a = 14.10$ and $b = -0.939$. The dashed gray line marks the 5σ candidate-detection threshold, crossed at $m_v = 13.30$ ($\Delta m = 7.30$ relative to the primary). The fitted limit has an uncertainty of approximately ± 0.1 mag.

Figure 4 compares 100- and 400-frame observations of an $m_v = +13.8$ companion. At 100 frames, the peak SNR is 3.4, below the candidate-detection threshold. At 400 frames, it reaches 9.4σ.

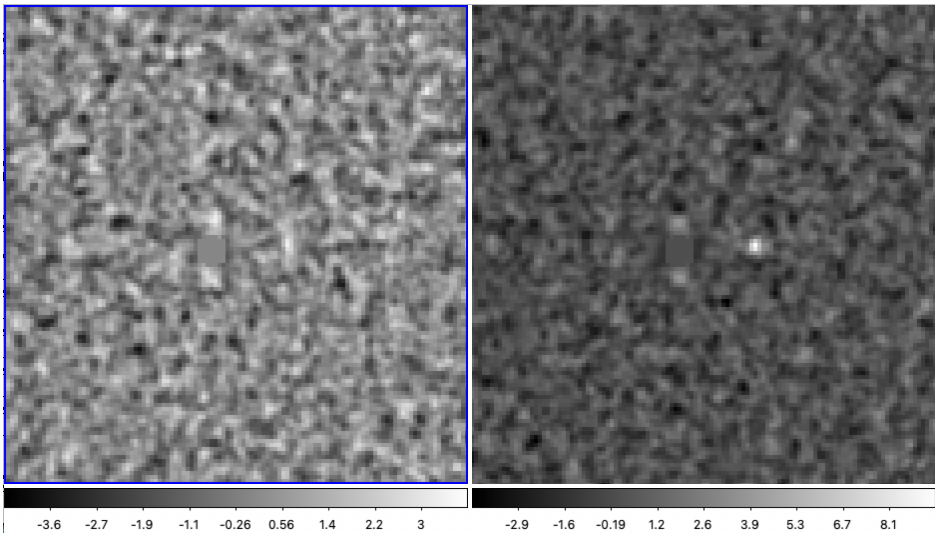


Figure 4: Matched-filter SNR maps for an $m_v = +13.8$ companion. Left: 100 frames; no credible companion peak is present. Right: 400 frames; the companion reaches 9.4σ. Residual signal at the primary location reflects imperfect recovery of the primary morphology. Both panels use logarithmic scaling.

This exceeds the factor-of-two gain expected from noise averaging alone. Across 100, 200, 400, and 800 frames, peak detection SNR follows

$$\mathrm{SNR} \propto N^{0.576}, \tag{10}$$

as shown in Figure 5. The excess over $\sqrt{N}$ arises because additional frames also improve the primary model and reduce the systematic residual pedestal. This benefit has a wavelength-dependent ceiling: spatial frequencies above the diffraction cutoff, $k_c \propto D/\lambda$, are absent regardless of frame count. At the red end of the bandpass, where k_c is low, reconstruction error is essentially unchanged from 100 to 400 frames. At the blue end, where k_c approaches Nyquist, it continues to decline (Figure 6). Thus, each wavelength has a saturation frame count $N^*(\lambda)$ beyond which additional frames provide little or no benefit.

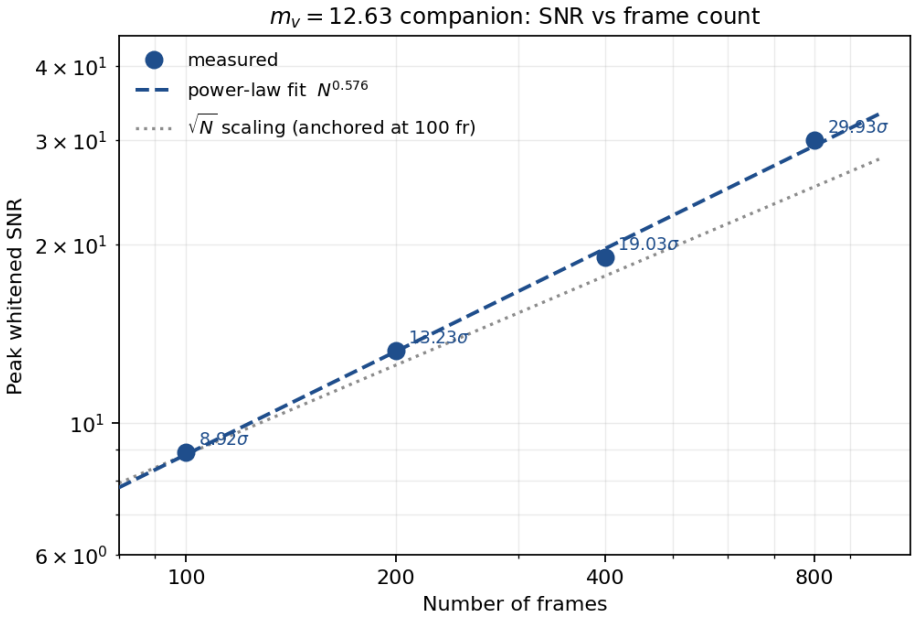


Figure 5: Peak detection SNR versus number of frames for the $m_v = +12.63$ companion. Circles show measured values at 100, 200, 400, and 800 frames. The dashed line is the fit SNR $\propto N^{0.576}$; the dotted line shows $\sqrt{N}$ scaling anchored at 100 frames.

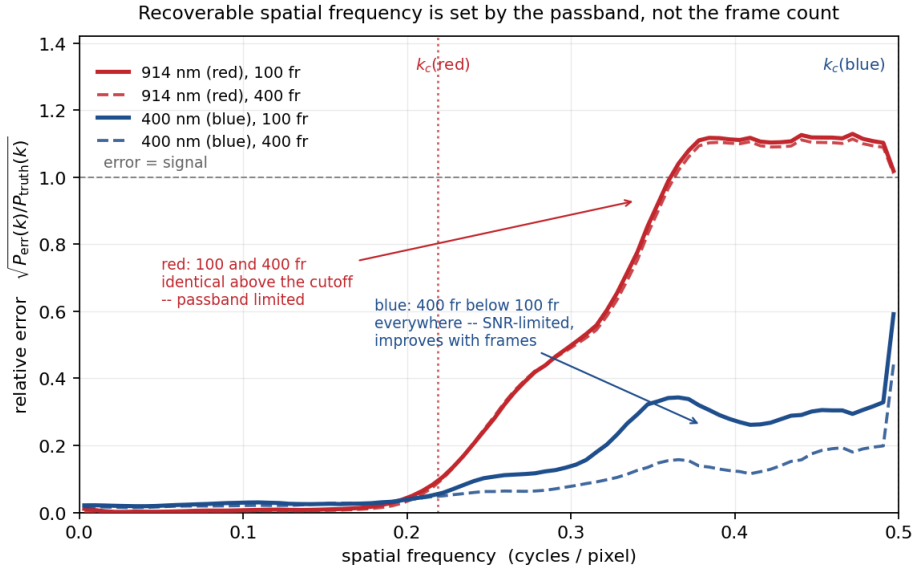


Figure 6: Recoverable spatial frequency for the reddest (914 nm) and bluest (400 nm) object slices at 100 frames (solid) and 400 frames (dashed). The vertical axis is $\sqrt{P_{\rm err}(k)/P_{\rm truth}(k)}$, the ratio of error to signal power in each spatial-frequency bin. Dotted vertical lines mark the diffraction cutoff, $k_c \propto D/\lambda$. The red reconstruction is support-limited and changes little with frame count; the blue reconstruction remains SNR-limited below the cutoff and improves as frames are added.

Joint linear regression for spectral characterization

The matched-filter map provides a finding chart, but detection alone does not ensure reliable spectral characterization. The relevant question is not only whether a companion can be detected, but whether its spectrum should be estimated from the restored image or from a joint fit to the original residual data. Figure 7 compares the true spectrum with that obtained by direct image restoration for the data in Figure 1; the two spectra show little similarity according to either fidelity metric.

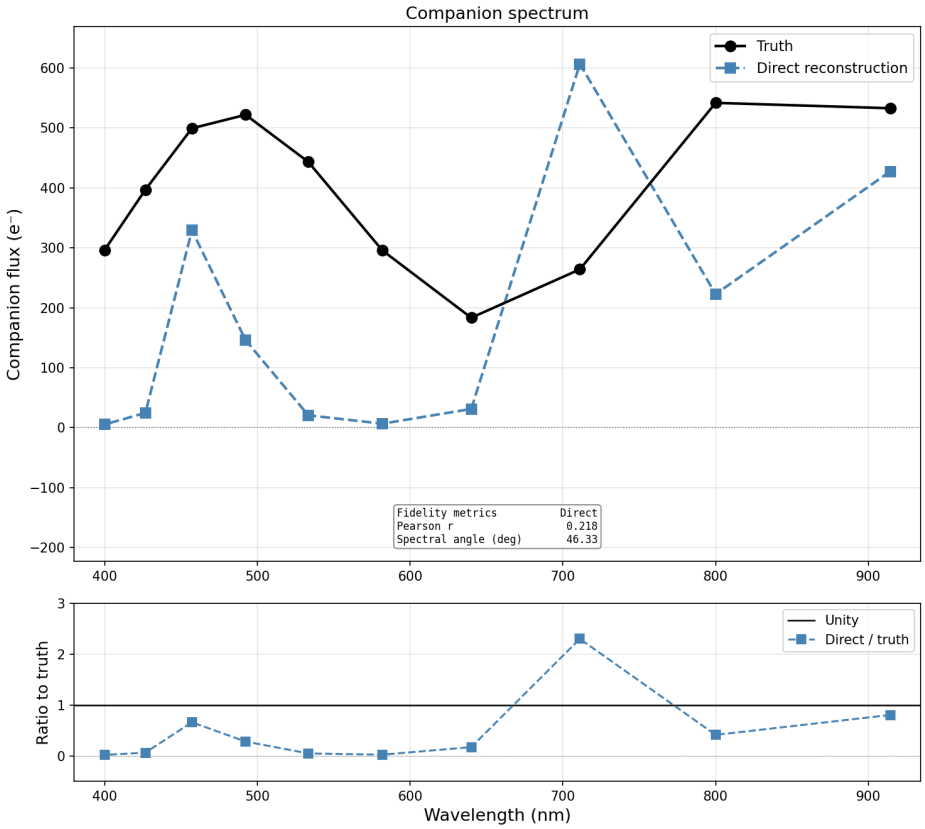


Figure 7: True companion flux and flux recovered by direct image restoration for the data in Figure 1. The direct flux estimate uses a 3×3-pixel aperture centered on the companion. Both spectral-fidelity metrics indicate little similarity between the true and recovered spectra.

Reliable material identification requires high spectral accuracy after accounting for transmission effects. We therefore estimate the companion spectrum using joint linear regression.

The companion contribution to the residuals in Equation 4 is

$$m_k^{r'}(\vec{x}) = \sum_{\lambda} \alpha_\lambda h_{\lambda,k}(\vec{x} - \vec{x}_0). \tag{11}$$

We estimate α_λ by weighted linear least squares within an aperture $\mathcal{A}$ centered at $\vec{x}_0$. Minimizing the weighted residual objective yields

$$\mathbf{A}\hat{\boldsymbol{\alpha}} = \mathbf{b}, \tag{12}$$

with solution

$$\hat{\boldsymbol{\alpha}} = \mathbf{A}^{-1}\mathbf{b}. \tag{13}$$

The vector $\mathbf{b}$ and matrix $\mathbf{A}$ are

$$b_\lambda = \sum_{k,\, \vec{x}\in\mathcal{A}} \frac{r'_k(\vec{x}) h_{\lambda,k}(\vec{x} - \vec{x}_0)}{\sigma_k^2(\vec{x})}, \tag{14}$$

and

$$A_{\lambda,\lambda'} = \sum_{k,\, \vec{x}\in\mathcal{A}} \frac{h_{\lambda,k}(\vec{x} - \vec{x}_0) h_{\lambda',k}(\vec{x} - \vec{x}_0)}{\sigma_k^2(\vec{x})}. \tag{15}$$

The vector $\mathbf{b}$ correlates residuals with PSFs shifted to the companion location and is analogous to the matched filter. The matrix $\mathbf{A}$ depends only on the PSFs, not on companion brightness. Its off-diagonal elements quantify spectral cross-talk caused by overlap among wavelength-dependent PSF templates.

Figure 8 shows the joint-regression spectrum for the $m_v = +13.1$ companion. Both fidelity metrics improve substantially relative to direct image restoration.

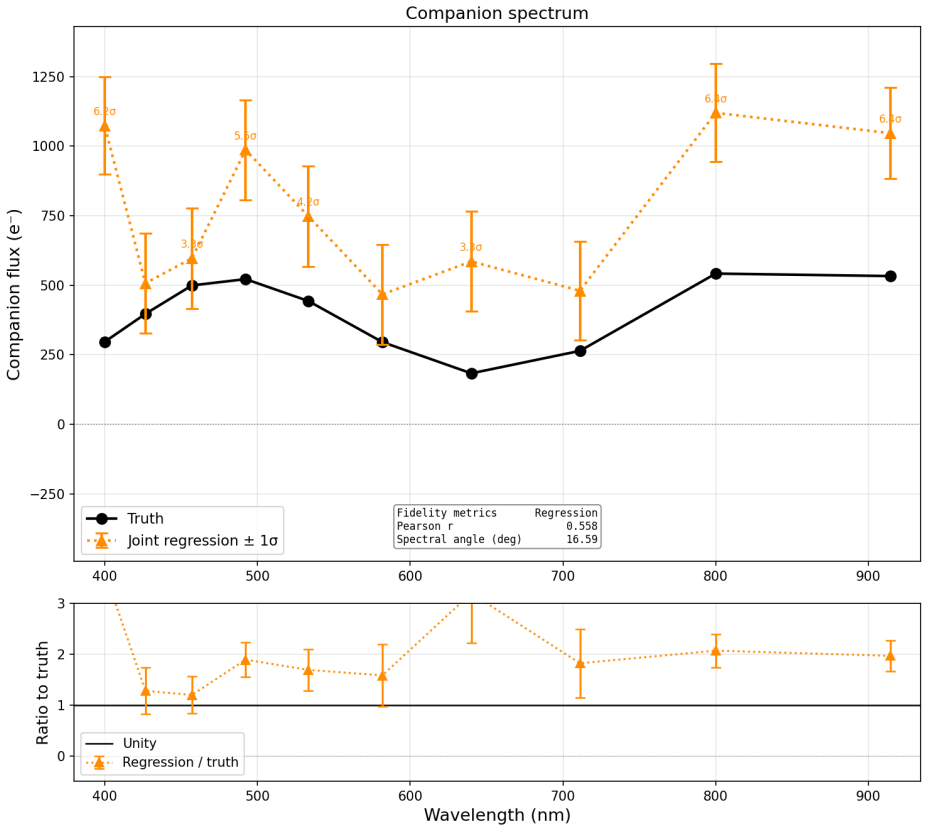


Figure 8: True companion flux and flux recovered by joint linear regression for the data in Figure 1. Both spectral-fidelity metrics improve substantially relative to direct image restoration in Figure 7.

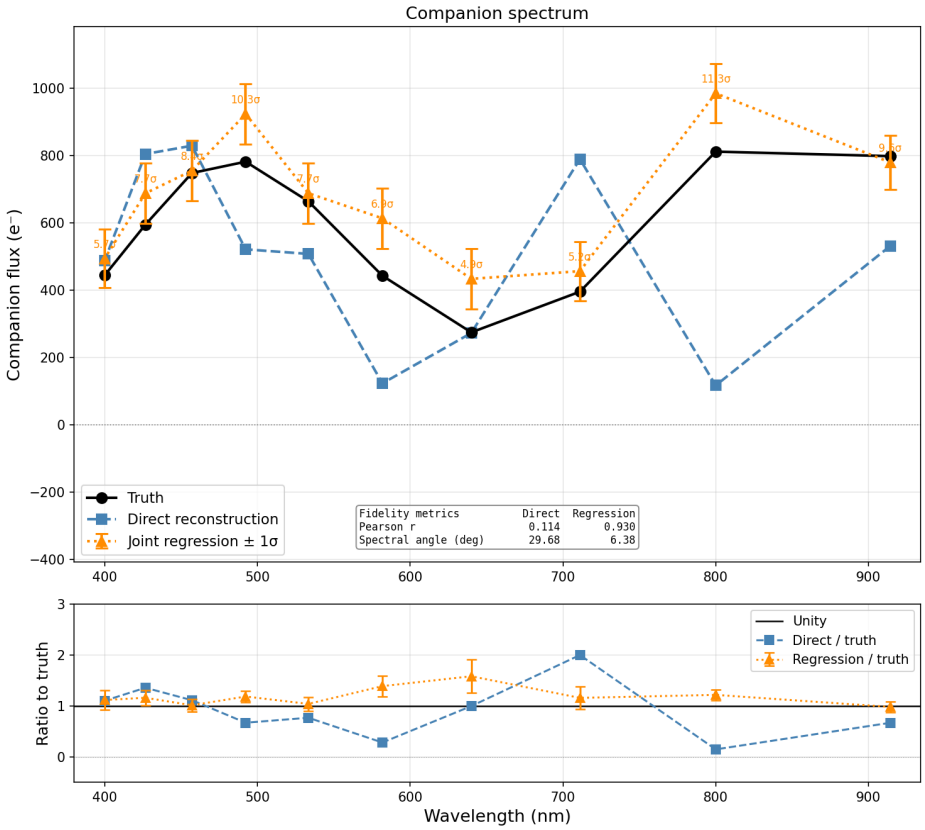


Figure 9: True companion flux, direct-restoration estimate, and joint-linear-regression estimate for 400 frames with an $m_v = +12.6$ companion. The companion has a peak matched-filter SNR of 19 and is readily detected in the direct restoration. Nevertheless, joint linear regression provides the more faithful spectral shape, showing that a high-SNR direct detection does not ensure reliable direct spectral characterization.

Figure 9 shows that the advantage of joint linear regression is not limited to marginal detections. In this 400-frame case, the $m_v = +12.6$ companion has a peak matched-filter SNR of 19 and is readily apparent in the direct restoration.

Nevertheless, the spectrum obtained by joint linear regression agrees more closely with the true spectral shape than does the spectrum extracted directly from the restored image. Thus, a high-SNR detection in the restored image does not by itself ensure reliable direct spectral characterization.

Joint linear regression uses the full residual data set and jointly fits the wavelength-dependent companion amplitudes while accounting for overlap among PSF templates. It therefore addresses spectral cross-talk explicitly, rather than treating reconstructed wavelength slices as independent flux estimates. It also provides uncertainties on the recovered fluxes, which are difficult to obtain from direct restoration without dividing the data into smaller subsets.



Figure 10: Spectral-shape fidelity versus peak detection SNR for 100-frame (blue circles), 200-frame (orange triangles), 400-frame (green squares), and 800-frame (purple diamonds) runs. Panel (a) shows spectral angle θ; panel (b) shows Pearson correlation r. Companion magnitudes are annotated. Dashed red lines mark the empirical good-shape criteria, $\theta < 10^\circ$ and $r > 0.8$; the dotted vertical line marks the approximate 14σ SNR threshold. Gray connectors link selected runs at fixed companion magnitude.

Figure 10 shows that spectral-shape fidelity is governed primarily by peak detection SNR rather than frame count. When plotted against SNR, results from all frame counts follow a common trend: spectral angle decreases and Pearson correlation increases with SNR.

Detection therefore requires substantially less SNR than reliable spectral characterization. In these simulations, good spectral fidelity from joint linear regression generally occurs for peak matched-filter SNRs above approximately 14σ, roughly three times the candidate-detection threshold.

Accomplishments in this reporting period

This reporting period produced four principal results:

1. We developed a radially whitened matched-filter statistic for locating unresolved companions after primary-satellite subtraction. In the modeled 100-frame observing scenario, the method reached an approximate 5σ candidate-detection limit of $m_v = +13.3$ for an $m_v = +6$ primary.
2. We found that detection SNR increased approximately as $N^{0.576}$ over 100–800 frames, exceeding the nominal $\sqrt{N}$ scaling because additional frames also improve recovery of the primary.
3. We identified a wavelength-dependent ceiling on this gain. Once the reconstruction reaches its diffraction- or support-limited regime, additional frames cannot recover spatial frequencies absent from the data and provide little or no further improvement.
4. We showed that joint linear regression can provide more faithful companion spectra than direct image restoration, including in cases where the companion is readily detected in the reconstructed image.

Current limitations

All performance results in this note are based on modeled data and the assumed PSF knowledge, primary morphology, brightness, and primary–companion separation. The reported detection and spectral-fidelity thresholds should therefore be treated as scenario-specific rather than as validated operational limits. In particular, on-sky PSF errors, temporal variability, unmodeled primary structure, and smaller separations may reduce performance.

Potential impact

If the simulated performance is retained on on-sky data, hyperspectral speckle imaging could provide a ground-based means to detect faint unresolved objects near brighter resident space objects and to estimate their spectral shape from broadband short-exposure data. The joint-regression result is particularly important because it suggests that reliable spectral characterization may be possible even when direct extraction from restored images is inadequate. Such capability could improve the identification and characterization of nearby debris or other uncooperative objects.

Next steps

The first priority is to test the methods on real on-sky data. During the next reporting period, we will acquire a suitable dataset to assess the effects of practical PSF uncertainty, temporal variability, primary morphology and brightness, and primary–companion separation.

We will also extend the joint spectral regression from a resolved primary with a faint unresolved companion to two unresolved point sources. This case is relevant to closely spaced GEO or cislunar objects observed with a 3-m-class ground-based telescope, where both objects are unresolved at the available angular resolution. A simultaneous two-source fit will estimate both spectra while accounting for overlap between wavelength-dependent PSF templates and the resulting flux covariance. This could enable identification and characterization of nearby objects—such as paired spacecraft, deployment-associated objects, or debris—whose spectra cannot be separated through conventional image restoration alone.

We will develop and test a two-point-source regression model that jointly estimates the wavelength-dependent amplitudes of both objects and then their relative positions. Simulations representative of GEO and cislunar observations with a 3-m-class telescope will evaluate spectral-recovery fidelity versus angular separation, brightness ratio, spectral contrast, frame count, and PSF error. The two-source template Gram matrix and inverse covariance will quantify blending and identify conditions under which both component spectra can be recovered reliably.

Lastly, we will investigate Maximum Residual Likelihood (MRL; Pina & Puetter 1992) as a way to reduce the correlated residual floor at fixed frame count, rather than relying solely on improved reconstruction from additional frames. In particular, we will evaluate our three-dimensional MRL implementation, which includes temporal as well as spatial lags. Temporal lags should suppress residual structure that persists across frames. The algorithm is operational (Appendix A.3), but its current implementation is too slow for practical use and requires optimization.

Appendix

A.1. Accelerated convergence

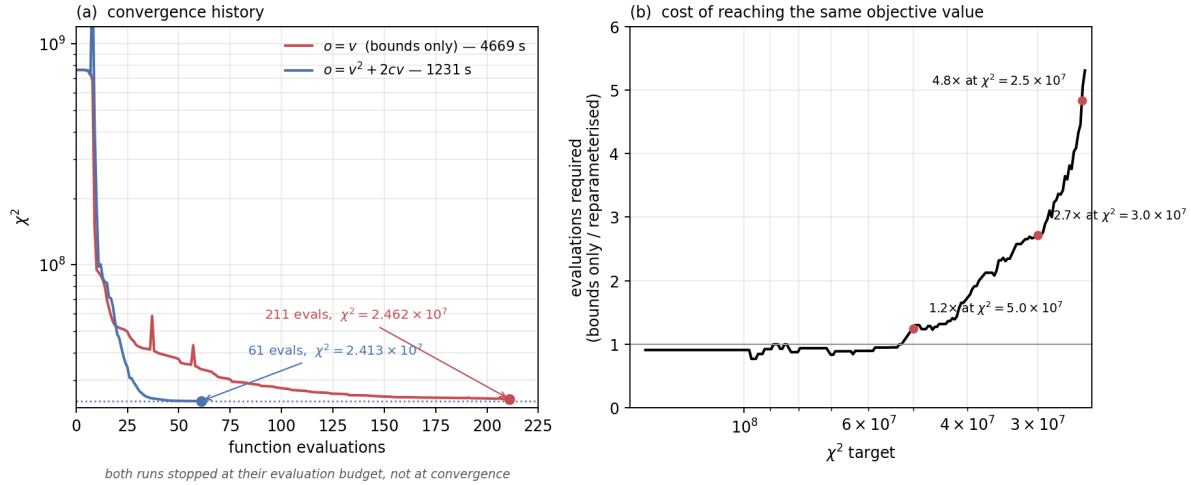


Figure 11: Effect of positivity parameterization on the optimization path for the $m_v = +12.63$ companion using 100 frames. Left: χ^2 versus L-BFGS-B function evaluations for $o = v$ with a zero lower bound (red) and for $o = v^2 + 2cv$, with $c = \sigma_{\mathrm{read}}\sqrt{g}$ (blue). Both parameterizations describe the same positive object values and therefore have the same attainable minimum; they differ only in their optimization path. The runs stopped at their evaluation budgets rather than convergence. On identical Apple M1 Max hardware, wall-clock times were 4668.9 s and 1230.5 s, respectively, a factor of 3.8. Right: ratio of function evaluations required by the bounds-only approach to those required by the reparameterization at a given χ^2.

L-BFGS-B enforces positivity of the recovered object, $o_\lambda(\vec{x}) = v(\vec{x})$, by imposing the bound $v(\vec{x}) \geq 0$. We find that the reparameterization

$$o(\vec{x}) = v(\vec{x})^2 + 2cv(\vec{x}), \qquad v(\vec{x}) \geq 0, \tag{16}$$

where c is determined by the camera read noise, reaches the same cost-function value as direct bound-constrained optimization in approximately one-quarter of the wall-clock time (Figure 11).

A.2. Spectral fidelity metrics

Let t_λ and $\hat{t}_\lambda$ denote the true and recovered companion fluxes in wavelength slice λ. Let $\mathcal{F}$ be the set of slices for which both values are finite, and let $\mathcal{V} \subseteq \mathcal{F}$ be the subset with $t_\lambda \neq 0$. Fractional metrics are evaluated over $\mathcal{V}$ and shape metrics over $\mathcal{F}$.

Pearson correlation coefficient.

$$r = \frac{\sum_{\lambda\in\mathcal{F}}(\hat{t}_\lambda - \bar{\hat{t}})(t_\lambda - \bar{t})}{\sqrt{\sum_{\lambda\in\mathcal{F}}(\hat{t}_\lambda - \bar{\hat{t}})^2}\sqrt{\sum_{\lambda\in\mathcal{F}}(t_\lambda - \bar{t})^2}}. \tag{17}$$

Here, $r = 1$ denotes perfect linear agreement in shape and $r = 0$ denotes no correlation.

Spectral angle.

$$\theta = \cos^{-1}\left(\frac{\sum_{\lambda\in\mathcal{F}}\hat{t}_\lambda t_\lambda}{\sqrt{\sum_{\lambda\in\mathcal{F}}\hat{t}_\lambda^2}\sqrt{\sum_{\lambda\in\mathcal{F}}t_\lambda^2}}\right), \tag{18}$$

reported in degrees. A value of $\theta = 0^\circ$ denotes identical spectral direction (Kruse et al. 1993).

Behavior. Both r and θ are invariant under $\hat{t}_\lambda \to c\,\hat{t}_\lambda$ for $c > 0$ and therefore measure spectral shape rather than absolute level. Because material identification depends primarily on reflectance shape, θ is the principal metric here. We use $\theta < 10^\circ$ and $r > 0.8$ as empirical criteria for good spectral-shape recovery (Figure 10).

The metrics are not equivalent: r subtracts the mean of each spectrum before correlation, whereas θ does not. A spectrum with the correct shape but a constant offset, $\hat{t}_\lambda = t_\lambda + \delta$, can therefore leave r unchanged while increasing θ.

A.3. 3-D Maximum Residual Likelihood estimation

Least-squares fitting minimizes total squared residual but does not constrain residual structure. Imperfect PSF estimates or unmodeled object features can leave spatially and temporally correlated residuals, violating the zero-mean, unit-variance Gaussian assumption required for matched filtering and spectral extraction.

We hypothesize that these correlations can be suppressed by extending the traditional Maximum Residual Likelihood (MRL) objective (Pina & Puetter 1992) to three dimensions. The extension includes temporal lags, τ, as well as spatial lags, $(\Delta x, \Delta y)$, in the autocorrelation function (ACF) of the whitened residuals:

$$\mathcal{L}_{\mathrm{MRL}} = \sum_{(\tau,\Delta y,\Delta x)\in\mathcal{S}}\left[\frac{C(\tau,\Delta y,\Delta x) - \mu_0(\tau,\Delta y,\Delta x)}{\sigma_0(\tau,\Delta y,\Delta x)}\right]^2, \tag{19}$$

where the three-dimensional sample ACF at lag $(\tau, \Delta y, \Delta x)$ is

$$C(\tau,\Delta y,\Delta x) = \frac{1}{L-\tau}\sum_{l=0}^{L-\tau-1}\sum_{\mathbf{x}\in\mathcal{M}}\tilde{r}_l(\mathbf{x})\tilde{r}_{l+\tau}(\mathbf{x}+\boldsymbol{\Delta}), \tag{20}$$

with

$$\tilde{r}_l(\mathbf{x}) = \frac{r_l(\mathbf{x})}{\sigma_l(\mathbf{x})}. \tag{21}$$

Here, $\mathcal{M}$ is an analysis mask defined as the object support dilated by the PSF radius, and $\boldsymbol{\Delta} = (\Delta y, \Delta x)$. The lag set $\mathcal{S}$ consists of half-plane spatial lags within radius R, together with zero-spatial-lag temporal lags up to $\tau_{\max}$:

$$\mathcal{S} = \left\{(\tau,\Delta y,\Delta x) : \tau = 0,\ \Delta y^2 + \Delta x^2 \le R^2,\ (\Delta y > 0 \text{ or } \Delta x > 0)\right\} \cup \left\{(\tau,0,0) : 1 \le \tau \le \tau_{\max}\right\}. \tag{22}$$

The null mean μ_0 and standard deviation σ_0 in Equation 19 are available in closed form. Because the lag set excludes $(0,0,0)$, whitened residuals with variance s^2 have

$$\mu_0 = 0, \qquad \sigma_0(\tau) = s^2\sqrt{\frac{N_{\mathcal{M}}}{L-\tau}}, \tag{23}$$

where $N_{\mathcal{M}}$ is the number of pixels in the analysis mask and L is the number of frames.

References

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